



# IoT Smart Pill Dispenser with Automated Medication Reminders Using Arduino UNO

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## KEYWORDS

Arduino UNO, Real-Time Clock (RTC), GSM Module, Embedded Systems, Remote Monitoring.

## ABSTRACT

Medication non-adherence is a major concern in healthcare, especially among elderly patients and individuals with chronic diseases. Missing or incorrect medication intake can lead to serious health complications. This paper presents an IoT based smart pill Dispenser using Arduino UNO that automates medication reminders and ensures timely and accurate dispensing. The system uses a Real-Time Clock (RTC) module for scheduling, a servo motor for automatic pill dispensing, and a buzzer with an LCD display for user alerts. A GSM module is included to send SMS notifications to patients or caregivers in case of missed doses, enabling remote monitoring. The proposed system offers a simple, reliable, and affordable solution to improve medication adherence, enhance patient safety, and support modern smart healthcare

## I. INTRODUCTION

In today's fast-paced lifestyle, maintaining proper medication schedules has become increasingly difficult, especially for elderly individuals and patients suffering from chronic diseases. Missing doses, taking incorrect medication, or consuming it at the wrong time can lead to serious health complications and reduced treatment effectiveness. Therefore, there is a strong need for an intelligent system that can assist patients in managing their medication routines efficiently.

With the rapid advancement of embedded systems and internet of things (IoT) technologies, healthcare devices are becoming smarter, more connected, and user

friendly. Automated medication systems have gained significant attention as they help reduce human errors and improve adherence to prescribed treatments [1]. IoT based solutions further enhance these systems by enabling real time monitoring and remote communication, which is particularly useful for caregivers and family members [2].

This project focuses on the development of an IoT based Smart Pill Dispenser using Arduino UNO, designed to automate medication reminders and dispensing. The system integrates key componenys such as a real time clock (RTC) module for accurate timing, a servo motor for controlled pill dispensing, an LCD

display for user interaction, and a buzzer for alert notifications. Additionally, a GSM module is incorporated to send SMS alerts in case of missed doses, ensuring continuous monitoring and support.

Similar systems proposed in earlier research emphasize cost effectiveness, ease of use, and reliability as essential features for practical healthcare applications [3][6]. By combining automation with IoT capabilities, the proposed system minimizes manual intervention, enhances patient safety and provides a dependable solution for medication management. Thus, this project demonstrates how low-cost embedded technology can be effectively utilized to address real-world health care challenges and improve overall quality of life.

## 2. LITERATURE SURVEY

These several research works have been carried out in the field of automated medication systems to improve patient adherence and reduce human errors. These studies provide a strong foundation for the development of smart pill dispensers using embedded systems and IoT technologies.

Karthika Hariprasad [1] proposed an Arduino based smart pill dispenser aimed at preventing drug overdose. The system uses a servo motor to dispense pills at scheduled intervals and highlights the importance of automation in ensuring medication safety and accuracy. The study emphasizes that such systems significantly reduce human intervention and errors.

Jyothis Philip et al.[2] developed an IoT based automatic medicine dispenser that focuses on improving medication adherence, especially for elderly patients. The system provides timely alert and dispenses pre-measured doses, reducing dependency on caregivers and enabling efficient medication management through connected technologies.

P. Ganga Bhavani et al. [3] introduced a smart medicine reminder box using Arduino UNO, RTC module, LCD display, and buzzer. The system provides both visual and audio alerts and allows customizable scheduling. It is designed to be user friendly, portable, and cost effective, making it suitable for everyday healthcare use.

Nikhil Gawali et al. [4] proposed an automatic pill dispenser machine to assist patients in managing complex medication schedules. The system ensures that medicines are dispensed at predefined times, reducing

the risk of missed doses and improving treatment outcomes.

Z. Nasir et al. [5] designed a smart medical box integrated with IoT features such as SMS notifications and biometric authentication. This system enhances security and monitoring, demonstrating how advanced technologies can improve patient care and medication tracking.

L.Gargioni et al. [6] presented a systematic review of various pill dispenser systems, highlighting their effectiveness in improving medication adherence and patient outcomes. The study discusses different technologies used in modern health care devices and emphasizes the growing importance of smart medical systems.

A. Jabeena et al. [7] proposed an automatic pill reminder system that reduces dependency on human memory. The system ensures correct dosage at the right time, thereby minimizing medication errors and improving overall health care management.

## 3. EXISTING METHOD

In current healthcare practices, medication management is primarily carried out using traditional methods such as manual pillboxes, alarm reminders, and caregiver supervision. These methods are simple and widely used but have several limitations in ensuring proper medication adherence.

One common method is the use of manual pill organizers, where patients sort their medicines based on days and timings. Although this helps in basic organization, it still relies heavily on the patient's memory and discipline. There is a high chance of missed doses or incorrect intake, especially among elderly individuals.

Another approach involves alarm-based reminders, such as mobile alarms or digital timers. These systems notify users when it is time to take medication, but they do not ensure whether the medicine has actually been taken. This lack of confirmation can lead to non-adherence or overdose.

In some cases, caregiver supervision is used, where family members or healthcare providers monitor the patient's medication schedule. While this method is more reliable, it increases dependency and is not always practical, especially for patients living independently.

Previous research has introduced basic automated systems that provide alerts and timed dispensing [1][4]. However, many of these systems lack advanced features such as remote monitoring, real-time notifications, and IoT connectivity. Additionally, some designs are complex or expensive, limiting their widespread adoption[6].

Existing methods are either manual, partially automated, or lack real time monitoring capabilities. These limitations highlight the need for a more efficient, automated, and connected solution for medication management, which is addressed by the proposed IoT based smart pill dispenser.

#### 4. PROPOSED METHODOLOGY

The proposed system is an IoT based Smart Pill Dispenser using Arduino UNO, designed to automate medication reminders and ensure accurate dispensing at scheduled times. The methodology focuses on integrating embedded hardware components with communication technology to create a reliable and user friendly healthcare solution.

The system is built around the Arduino UNO microcontroller, which acts as the central control unit. A Real time Clock (RTC) module is used to maintain precise timing for medication schedules. Users can set the required dispensing times using input switches, making the system easy to configure according to individual needs.

When the preset time is reached, the Arduino processes the signal from the RTC and activates a servo motor mechanism to dispense the required number of pills automatically. At the same time, a buzzer alert is triggered, and a message is displayed on the LCD screen to notify the user about the medication. This ensures both visual and audio confirmation.

To Enhance system reliability, a GSM module is integrated for communication. If the patient does not take the medicine within a specified time, the system sends an SMS notification to caregivers or family members. This feature enables remote monitoring and reduces the risk of missed doses.

This methodology combines automation, real time monitoring, and IoT communication to overcome the limitations of existing systems. Compared to earlier approaches that mainly focus on reminders or basic automation [2][3], the proposed system provides an

integrated solution with improved accuracy, safety, and user convenience.

#### BLOCK DIAGRAM & WORKING SYSTEM

Block Diagram:

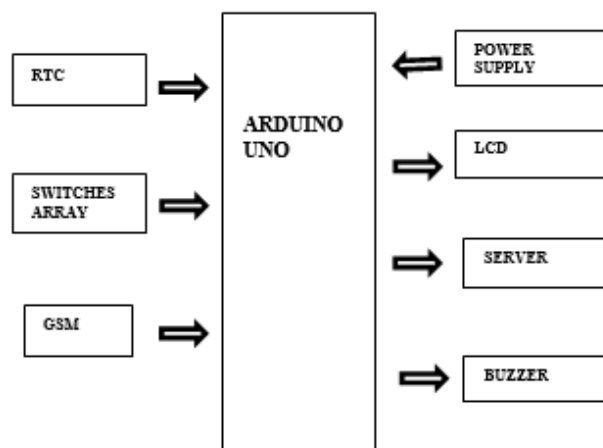


Fig 4.1: BLOCK DIAGRAM

Working System:

The working of the proposed IoT based Smart Pill Dispenser is based on the coordination of different hardware components controlled by the Arduino UNO. The system operates in a step by step manner to ensure timely medication dispensing and proper user notification.

Initially, the user sets the medication schedule using input switches. These settings are stored in Arduino, while the Real Time Clock (RTC) module continuously keeps track of the current time. The RTC ensures high accuracy in maintaining the schedule even during power interruptions.

When the current time matches the present medication time, the Arduino triggers the system. A Servo motor is activated to rotate the dispensing mechanism and release the required pills. At the same time, a buzzer alert is generated, and the LCD display shows a message indicating that it is time to take the medicine. This provides both audio and visual notification to the user.

If the user takes the medication, the system resets and waits for the next scheduled time. However, if the medication is not taken within a certain period, the GSM module is activated. It sends an SMS alert to caregivers or family members, informing them about the missed

dose. This ensures continuous monitoring and timely intervention.

The system continues this cycle for all scheduled timings throughout the day. The integration of automation and communication and communication makes the process reliable and efficient compared to traditional methods [2][4]. The working system ensures accurate time-based dispensing, immediate alerts, and remote monitoring, improving adherence and overall patient safety.

## 5. RESULTS AND OUTCOMES

The implementation of the IoT based Smart Pill Dispenser using Arduino UNO demonstrates effective performance in automating medication management and improving adherence. The system was tested under different conditions to evaluate its reliability, accuracy, and usability.

The results show that the Real Time Clock (RTC) module provides precise timing, ensuring that medicines are dispensed exactly at the scheduled intervals. The servo motor mechanism operates accurately to deliver the correct dosage without failure. The combination of buzzer alerts and LCD display successfully notifies users, making the system easy to understand and operate, even for elderly individuals.

The integration of the GSM module proved highly effective for remote monitoring. In cases where doses were missed, SMS notifications were sent promptly to caregivers, allowing timely intervention. This feature significantly enhances patient safety and reduces the risk of missed medication.

Compared to traditional and existing systems, the proposed model shows improved efficiency by combining automation with communication technology [2][6]. It reduces human errors such as missed doses and incorrect timing, while also minimizing dependency on caregivers. Thus, the developed system successfully achieves its objective of improving medication adherence and contributing to smarter healthcare solutions.

## 6. CONCLUSION

The proposed IoT based Smart Pill Dispenser using Arduino UNO provides an effective solution to the problem of medication non adherence. By integrating components such as the RTC module, servo motor, LCD display, buzzer, and GSM module, the system

successfully automates both medication reminders and dispensing processes.

The project demonstrates how embedded systems and IoT technologies can be applied in healthcare to improve patient safety and treatment effectiveness. The automated mechanism ensures that medicines are taken at the correct time and in the right dosage, thereby reducing human error such as missed doses or overdosing. The GSM based notification feature further enhances reliability by enabling remote monitoring and timely intervention by caregivers.

Compared to existing methods, the proposed system offers a more efficient, user-friendly, and cost-effective approach to medication management [2][6]. It reduces dependency on manual supervision and provides a practical solution for elderly patients, individuals with chronic illnesses, and busy users.

This smart pill dispenser represents a significant step toward modern health care solutions by combining automation, accuracy, and connectivity. It has the potential to improve quality of life, ensure better medical compliance, and serve as a foundation for future advancements in smart healthcare systems.

## Conflict of interest statement

Authors declare that they do not have any conflict of interest.

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